

LED Greeting Card

My Glowing Diwali Card Adventure!



Presented by
PLANRS Academy Master Course

Let's Light Up Diwali!

Diwali is the Festival of Lights! [cite: 455] It is a time when we celebrate with our families and decorate our homes with bright lamps.

Just like tiny clay Diyas glow with oil and wick, today we will make a card that glows using Electricity! [cite: 456]

Electricity = Bijli

Bijli is the magic energy that will light up our card! [cite: 457]

Are you ready to become a little scientist and artist at the same time? Let's go!

Meet Our Circuit Friends

Just like you need friends to play, our light needs three special friends to work together! [cite: 468]

1. Battery = "Fatty"

The Power House! It gives all the energy to our card, just like an inverter at home during a power cut. [cite: 459, 460, 477]

2. LED = "Leggy"

Our Little Diya! This tiny bulb uses electricity instead of oil to glow bright. [cite: 463, 464, 482]

3. Copper Tape = "Shiny Path"

The Bijli ki Taar! It is like the electric wires on the street that carry power like magic roads. [cite: 465, 466, 490, 492]

The Secrets of Leggy & Fatty

To make our circuit work, we must know the "secret sides" of our friends! [cite: 478]

Leggy's Long and Short Legs:

Look closely at your LED. One leg is longer than the other! [cite: 485, 486]

- *Long Leg: The Positive side (+) [cite: 485]*
- *Short Leg: The Negative side (-) [cite: 486]*

Fatty's Positive and Negative:

The Battery also has a "Positive" and "Negative" side. [cite: 478]

Important Scientist Rule: We must match the Long Leg (+) to the Positive side to make the light glow! [cite: 503]

Copper Tape: The Shiny Path

*Our Copper Tape is a special path for **Bijli** to travel. [cite: 491, 493]*

Imagine the electricity is like a tiny car. It can only drive on a perfect road. [cite: 493]

 *Scientist Warning: If the tape breaks or has a gap, the path is broken and the light won't glow! [cite: 494] Keep your Shiny Path perfect!*

We will stick the tape on our card to connect Fatty the Battery to Leggy the LED. [cite: 498, 501]

Guided Build - Step by Step

Follow these easy steps to make your card glow! [cite: 507]

Step 1: Stick the copper tape paths on the card carefully. [cite: 498, 499]

Step 2: Place the battery on the tape—Fatty is home! [cite: 501]

Step 3: Attach Leggy's legs on the tape and make a Fold-Switch! [cite: 503]

When you fold your card, watch Leggy glow bright! [cite: 508]

Decorate Your Card!

Now make your card beautiful with Indian motifs! [cite: 523, 524]

- Rangoli Art: Draw patterns around your tape paths. [cite: 512, 513]
- Diya Decor: Add colorful stickers or drawings around the LED. [cite: 517, 518]
- Festive Greetings: Write "Happy Diwali," "Eid Mubarak," or "Happy Birthday!" [cite: 519, 520, 521]



"Every light you make brightens the world around you!" [cite: 543]